



Spatial Modeling of Sugarcane Yield Using Machine Learning Approaches

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Abstract

Sustainable precision agriculture and artificial intelligence (AI) are transforming modern farming by enhancing resource efficiency, addressing climate challenges, and fostering sustainability. In Colombia, sugarcane cultivation is pivotal in the agricultural economy, particularly in the Cauca River Valley, a region grappling with climate variability and resource optimization challenges. This study introduces an innovative methodology that combines machine learning (ML) techniques with geospatial data to predict sugarcane yields accurately. The research employs the CatBoost, random forest, and XGBoost ML algorithms, integrating historical data on climate, soil characteristics, and agricultural management practices to develop spatially detailed predictive models. The findings demonstrate that ML-based approaches outperform traditional methods like linear and penalized regressions, offering more precise agricultural planning and resource management predictions. Focused on the municipality of La Candelaria, this work underscores the potential of ML technologies to enhance productivity while promoting sustainable agriculture.

Keywords Sugarcane · Machine learning · Spatial prediction · Water resources · Sustainable agriculture · Yield estimation

Introduction

Sustainable precision agriculture (Zhang et al. 2021) and advancements like artificial intelligence (AI) are revolutionizing sugarcane production by improving resource

efficiency, addressing climate challenges, and fostering sustainable development (Amiri et al. 2024; Shaamala et al. 2024; Su et al. 2023; Batool et al. 2022). These innovations are particularly relevant for optimizing sugarcane yield, a crop of significant global importance. AI-powered systems have been shown to increase crop yields by up to 30% while reducing water usage by 20–50% (Muqaddas et al. 2024).

Colombia, one of the world's top sugarcane producers, relies heavily on this sector, contributing substantially to its agricultural and industrial gross domestic product (GDP) (Naciones Unidas 2021). Concentrated in the Cauca River Valley, sugarcane cultivation spans 240,000 hectares, supports small- and medium-scale farmers, and plays a vital role in local employment and economic growth (Asocaña 2017). The region's favorable climate and soil conditions make it ideal for sugarcane farming. At the same time, the sector also drives exports of sugar and ethanol, further boosting Colombia's economy and global trade presence.

Within this context, accurate prediction of sugarcane yield is increasingly vital for ensuring food security, economic stability, and sustainable agricultural practices, especially in the face of climate variability and resource constraints (Sridhara et al. 2024). Recent literature has

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demonstrated that traditional statistical approaches, while interpretable, often fail to capture the complex, nonlinear interactions among climatic, soil, and management covariates that drive crop productivity (Shahhosseini et al. 2019).

Machine learning (ML) techniques have emerged as powerful tools for yield prediction due to their ability to process large, heterogeneous datasets and model intricate relationships between environmental factors and crop outcomes (Kheir et al. 2024). For instance, studies have shown that ML models such as random forest, artificial neural networks, and ensemble methods consistently outperform conventional regression models in forecasting sugarcane yields, achieving higher predictive accuracy and robustness across diverse agroecological zones (Sridhara et al. 2024). Notably, rainfall, temperature, and relative humidity have been identified as key determinants of yield, with ML models effectively integrating these covariates for improved forecasting.

Furthermore, the integration of explainable AI techniques has enhanced the interpretability of ML predictions, enabling actionable insights for precision agriculture and climate adaptation strategies (Mohan et al. 2025). These advancements not only support data-driven decision-making at the farm and policy levels but also facilitate adaptive management practices to mitigate the adverse impacts of climate change on sugarcane production.

This study leverages ML techniques (Hu et al. 2024; Bali and Singla 2022) and geospatial data (Breunig et al. 2020) to develop a robust framework for predicting sugarcane yields with high accuracy. The proposed methodology employs the CatBoost (Hancock and Khoshgoftaar 2020), random forest (Parmar et al. 2018), and XGBoost (Bentéjac et al. 2021) ML algorithms, integrating historical data on climate, soil properties, and agricultural management practices to generate spatially detailed yield estimates. These predictive models aim to optimize resource allocation, such as water and nutrients, while addressing the challenges of climate variability. Focused on the Cauca River Valley, particularly the municipality of La Candelaria, this approach offers a data-driven solution to enhance agricultural planning and promote sustainable sugarcane production. Given these developments, this study addresses the pressing need for reliable, scalable, and interpretable yield prediction frameworks in sugarcane cultivation, contributing to the growing body of research advocating for the adoption of advanced data-driven methodologies in sustainable agriculture.

By integrating geospatial data layers and ML-derived covariate importance analysis, the study identifies seasonal irrigation, minimum relative humidity, and solar radiation as the most influential factors affecting sugarcane yield in the study area. These findings provide actionable insights for targeted water management and adaptive agronomic strategies to improve yield and sustainability under local climate variability.

Although Colombian research has advanced digital agriculture—for example, Arrubla-Hoyos et al. (2022) deployed 5 G-enabled sensors in the Cauca Valley sugarcane region for real-time monitoring, and Maldaner et al. (2024) conducted spatial–temporal statistical analyses of sugarcane yield—these approaches rely mainly on sensing technologies or traditional statistical frameworks. Sectoral reports from Asocaña (Colombian Association of Sugarcane Producers), (2023); Asocaña (Colombian Association of Sugarcane Producers) (2024) and Agronet (2024) provide valuable production statistics and insights into agronomic practices but do not address predictive modeling with ML. To the best of our knowledge, no published studies in Colombia have applied spatial ML models for sugarcane yield prediction. Thus, this study provides one of the first contributions in this direction by integrating geospatial data with ensemble ML algorithms to generate spatially explicit yield estimates.

Previous studies on sugarcane yield prediction and related crops have largely relied on statistical modeling frameworks due to their interpretability and well-established theoretical foundations. Spatial and regional regression approaches have been widely used to relate sugarcane yield to climatic, soil, and management covariates, enabling the quantification of individual and combined environmental effects on crop productivity (Maldaner et al. 2024). Classical multivariate linear regression remains a common choice in agronomic applications because of its robustness and transparency, facilitating environmental impact assessment and decision-making. For instance, Viana et al. (2023) successfully applied multiple linear regression to estimate sugarcane yield in subtropical regions using climatic variables and soil water storage, demonstrating the ability of regression-based models to capture yield dynamics across growth stages. In parallel, remote sensing approaches have gained prominence by integrating satellite-derived indicators such as evapotranspiration and vegetation indices to monitor crop health and estimate yield at regional scales. An example is the agrometeorological framework proposed by Tripathy et al. (2023), which combined evapotranspiration and NDVI to predict sugarcane yield across multiple districts in India, achieving strong agreement between predicted and observed values.

Despite their advantages, traditional statistical and remote sensing-based methods often struggle to represent the complex, nonlinear interactions and fine-scale spatial variability that characterize sugarcane production systems. Time series and regression-based techniques, while valuable for interpretability and inference, may exhibit limited predictive performance when environmental drivers interact in nonadditive or heterogeneous ways (Mulianga et al. 2013). Recent advances in machine learning offer an opportunity to overcome these limitations by modeling nonlinear relationships and high-dimensional interactions among climatic, soil, and management variables. Ensemble tree-based methods,

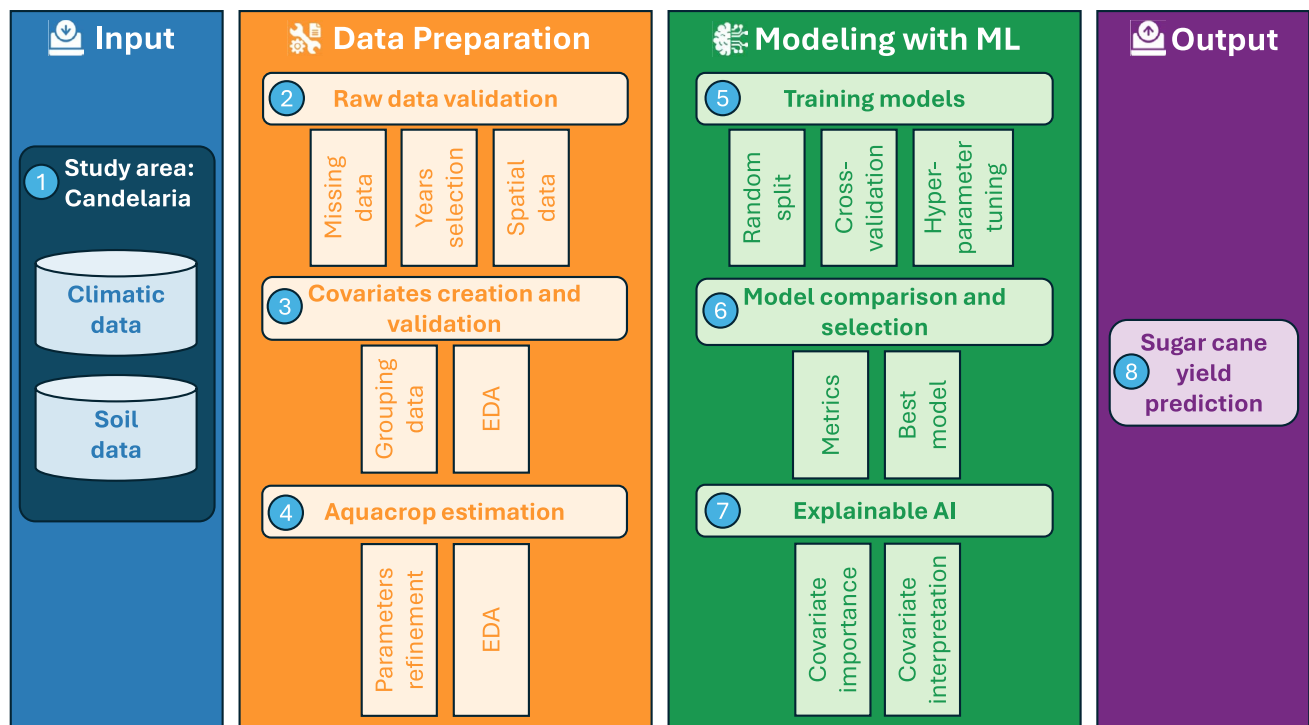


Fig. 1 Conceptual AI-powered methodology used for sugarcane yield prediction

in particular, have shown improved predictive accuracy while retaining a degree of interpretability through covariate importance analysis. Building on this evolution, the present study develops a spatial machine learning framework based on random forest, XGBoost, and CatBoost models to improve sugarcane yield prediction, benchmark performance against traditional statistical approaches, and identify the most influential drivers of spatial yield variability using model-derived importance metrics.

Materials and Methods

ML formulation and model interpretability

Sugarcane yield prediction was formulated as a supervised regression problem (Kopczewska 2022; Shetty et al. 2022), where ensemble tree-based ML models (random forest, XGBoost, and CatBoost) were trained to learn the nonlinear relationship between environmental covariates (climatic, soil, and management variables) and sugarcane yield. Model training aimed to minimize prediction error, quantified using the root mean squared error (RMSE), following standard practice in agronomic modeling and ML applications (Karusasingha 2022; Hodson 2022).

Beyond predictive accuracy, model interpretability was explicitly addressed to identify the main drivers of spatial

yield variability. For this purpose, covariate importance was assessed using permutation importance, a model-agnostic approach that quantifies the decrease in predictive performance when the values of a given covariate are randomly permuted (Bentéjac et al. 2021; Chicco et al. 2021). Unlike impurity-based importance measures commonly used in tree ensembles, permutation importance provides a more robust assessment of covariate influence and reduces biases associated with variable scale or cardinality.

A detailed mathematical formulation of the learning objective and the permutation-based importance computation is provided in Supplementary Material.

Methodology

This study utilizes an ML-based spatial modeling approach to predict sugarcane yield in Candelaria, Valle del Cauca, Colombia. The methodology integrates geospatial, climatic, and agronomic data within an ML framework to enhance yield prediction accuracy. The code developed for this study is available in the corresponding GitHub repository.¹

Figure 1 presents the improved methodology, organized into four blocks that mirror the conceptual pipeline in the

¹ GitHub repository: <https://github.com/dsocadaguic/PrediccionCanaAzucarAA>

figure: *input, data preparation, modeling with ML, and output*. The workflow proceeds through eight numbered steps: (1) study area and data sources; (2) raw data validation; (3) covariates creation and validation; (4) AquaCrop estimation; (5) model training; (6) model comparison and selection; (7) explainable AI; and (8) sugarcane yield prediction.

In the following, we detail each stage in accordance with Fig. 1.

Input (Step 1)

Study area

The study area chosen for this work is depicted in Fig. 2. It corresponds to Candelaria in southwest Valle del Cauca, Colombia, which covers approximately 266 km² at 1,000 meters above sea level. It has a tropical humid climate, with an annual average temperature of 24°C and a bimodal rainfall pattern averaging 1,200 mm (Asocaña 2017; Agronet 2024). Soil characteristics, such as a slightly acidic pH (5.5–6.5) and medium to high organic matter content (3–6%), make it suitable for sugarcane cultivation. Candelaria contributes

significantly to regional sugarcane production due to these favorable agroecological conditions (Arias *n.d.*).

Climatic data

Cenicaña supplied a daily series of maximum and minimum temperatures, precipitation, solar radiation, relative humidity, and reference evapotranspiration from a network of 15 meteorological stations in and around Candelaria (records spanning 2010–2022). For downstream modeling, we used the complete daily year 2022. To comply with AquaCrop input requirements, station series were converted into .CLI, .TMP, .PLU, and .ETO files, ensuring full coverage of the sugarcane cycle without gaps.

Soil data

Cenicaña and IGAC provided detailed soil information, including texture, pH, organic matter, and macronutrient contents (N, P, K). For AquaCrop simulations, soil profiles were parameterized with key physical properties such as texture, field capacity, and permanent wilting point. These

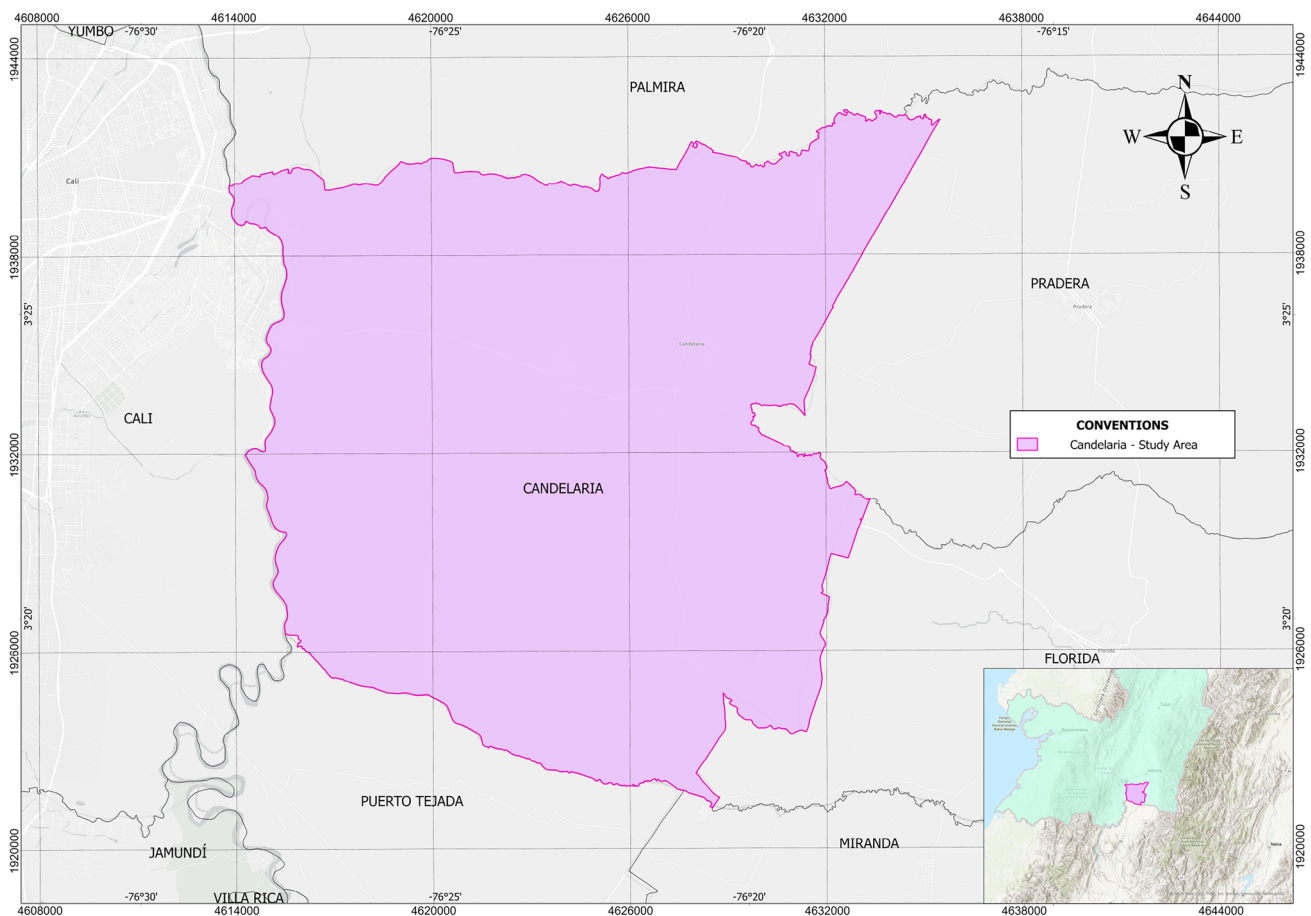


Fig. 2 Study area showing its position in Colombia and limiting regions

values were assigned to each cartographic unit using modal soil profiles, later rasterized and aligned with the climatic layers to generate pixel-level soil attributes for the study area.

Data Preparation (Steps 2–4)

Raw data validation

Daily climatic series (2010–2022) were audited for completeness and consistency (missing data screening, removal of incomplete years), and the spatial domain was restricted to the study area. While the network comprises 15 stations, 12 stations fell within or directly adjacent to the analysis grid for spatialization; these include *Meléndez*, *PTAR Cali*, *Candelaria*, *El Tiple*, *Ortigal*, *Bocas de Palo*, *Cenicaña*, *Pradera*, *Florida*, *Miranda*, *Aeropuerto*, and *Palmira San José*. All spatial processing followed standard geospatial practices and is detailed in Supplementary Material.

Covariates creation and validation

From the original daily climatic variables, a set of derived covariates was constructed by aggregating minimum, maximum, and mean values over the full crop cycle and the final growth stage. Soil type and seasonal irrigation were added, yielding a total of 44 predictors used for model training. Table 1 shows a sample of the consolidated dataset.

AquaCrop estimation

Field-level sugarcane yield data for La Candelaria are not publicly available due to confidentiality agreements with the Cenicaña research center and local sugar mills, including Mayagüez (2023), Providencia and (2025), and Castilla (2025). Accordingly, sugarcane yield was generated using the AquaCrop crop simulation model (version 7.1; Food and Agriculture Organization—FAO), which estimates crop productivity based on water use efficiency and environmental conditions.

AquaCrop simulations were automated using the AquaCrop-OS Python library, generating 1,000 yield realizations under varying climatic, soil, and management conditions, including irrigation and groundwater depth, and covering harvest cycles ranging from 4 to 12 months.

The use of AquaCrop-simulated yields in data-scarce contexts is supported by prior studies demonstrating its reliability and adaptability. Coudron et al. (Coudron et al. 2023) highlighted its robustness under limited empirical data, Garcia-Vila et al. (2019) demonstrated its effectiveness in water-constrained environments, and Sajid et al. (2022) showed that integrating crop simulation models with machine learning is a well-established strategy for capturing causal relationships between environmental drivers and yield.

Modeling with ML (Steps 5–7)

Training models

Random forest, XGBoost, and CatBoost models were trained using a supervised learning framework. Model optimization relied on cross-validation (CV) (King et al. 2021) and hyperparameter tuning (Morales-Hernández et al. 2023) to minimize RMSE. Categorical variables were encoded as needed, while continuous variables were used without scaling due to the tree-based nature of the models.

Model comparison and selection

Models were compared using RMSE, NRMSE (Moriasi et al. 2007), and R^2 (Chicco et al. 2021). The model with the best mean CV-RMSE was refitted on the full training data and evaluated on a held-out test set; the selected model was then used for spatial prediction. This procedure ensured that the final model combined both statistical accuracy and robustness before being deployed for yield mapping.

Explainable AI

Model interpretability was assessed using permutation-based feature importance to identify the main drivers of spatial yield variability. The formal definition and implementation details are provided in Supplementary Material.

Experimental Setup

The main aspects of the experimental design for the methodology to be reproducible are depicted in Supplementary Material.

Table 1 Sample of the dataset used

ID	Latitude	Longitude	Precip. (mm)	T. Min (°C)	T. Max (°C)	Soil Texture
1	3.4082	−76.3560	1345	19.3	32.8	Clay Loam
2	3.4156	−76.3705	1289	20.1	31.5	Silty Clay
3	3.4268	−76.3421	1412	18.9	33.2	Sandy Loam

Output (Step 8)

The best model was implemented to generate spatially explicit sugarcane yield predictions across Candelaria, enabling the identification of high- and low-yield zones and supporting decision-making in precision agriculture. The resulting maps were compared with AquaCrop outputs to highlight areas of agreement and discrepancy. This comparison is intended as an internal consistency check between the machine learning predictions and the AquaCrop simulations, rather than as an independent validation exercise.

Results

Sugar cane yield exploratory analysis

Our study analyzed the distribution of estimated sugarcane yield in T/ha using a histogram and a violin plot as shown in Fig. 3. The histogram revealed that, on average, sugarcane yields reach between 79 and 109 T/ha, with the 94–109 T/ha range being the most frequent, while yields below 49 T/ha were rare, indicating that low-yield plots are uncommon. The violin plot provided a detailed distribution view, combining a box plot with density visualization. It showed the highest density of yields between 70 and 100 T/ha, with a median close to 90 T/ha. The data exhibited a slight positive skew, suggesting some plots had higher-than-average yields,

extending the distribution's right tail. Importantly, no outliers were detected in the yield data.

Covariates exploratory analysis

The descriptive statistics for meteorological and soil covariates are shown in Table 4 of Supplementary Material, used as inputs for the sugarcane yield prediction model, including metrics such as mean, standard deviation, minimum and maximum values, quartiles, and the percentage of missing data. Minimum temperature (TempMin) averages 15.21°C with low variability (standard deviation of 0.91°C), indicating consistent nighttime conditions. In comparison, maximum temperature (TempMax) averages 27.34°C with more significant variability (standard deviation of 1.57°C), which may signal potential heat stress during warmer periods. These temperature ranges are critical for understanding the physiological development of the crop.

Precipitation shows significant variability, with an average of 1.91 mm and a standard deviation of 3.96 mm, ranging from 0 mm to 26.86 mm. This irregularity highlights periods of drought and heavy rainfall, which can impact sugarcane growth. Evapotranspiration averages 7.83 mm, ranging from 4.60 mm to 10.10 mm, reflecting water loss to the atmosphere that must be replenished to maintain soil moisture balance. Solar radiation (RadSolar), essential for photosynthesis, averages 468.55 with a standard deviation of 23.63, showing relative stability but occasional fluctuations

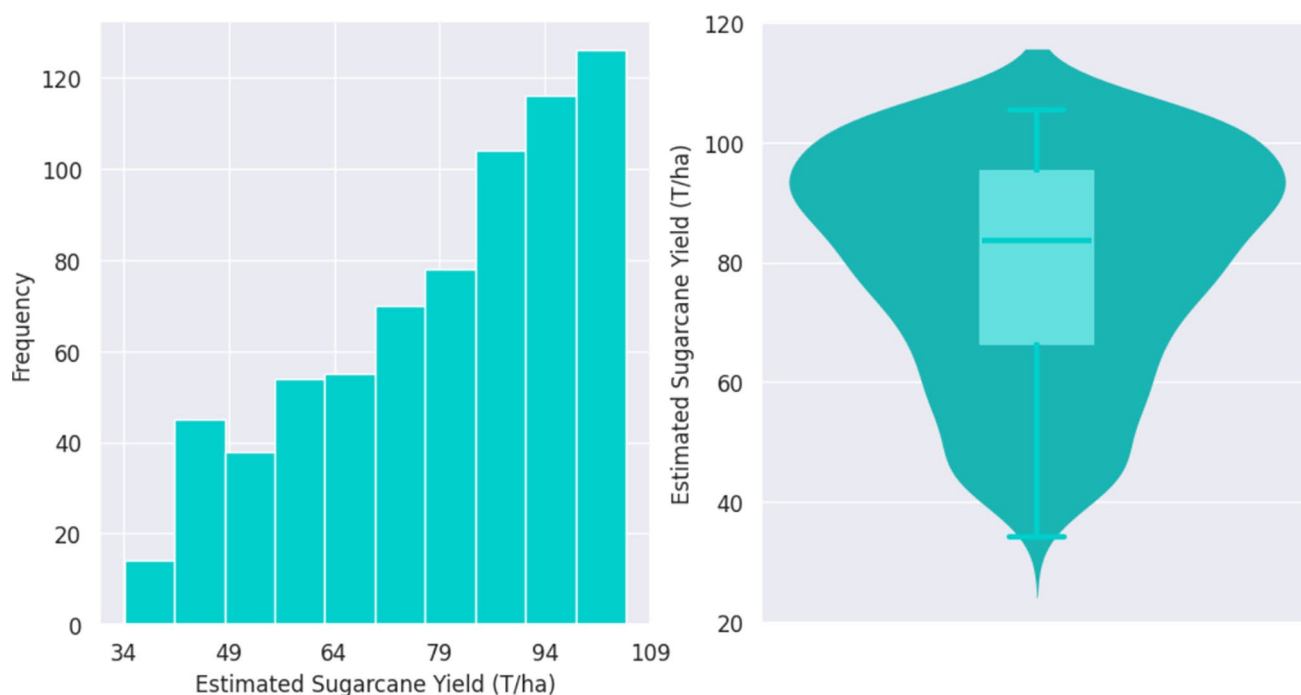


Fig. 3 (Left) Sugarcane yield histogram, (Right) sugarcane yield violin plot

between 387.78 and 508.96 that could influence biomass production.

Soil types, including clayey, silty clay loam, clay loam, and sandy loam soils, are represented as binary covariates indicating their presence in different areas. These soil categories are critical for water retention and nutrient availability but exhibit limited variability (90% of values are zero), which may reduce their predictive utility in ML models. Their uniformity could constrain the ability to capture differences across cultivation zones, potentially affecting model performance in identifying key factors influencing yield variability.

This study used Spearman's correlation to analyze non-linear but monotonic relationships between covariates (see Fig. 7 in Supplementary Material). The analysis revealed that seasonal irrigation has a moderate positive correlation (0.35) with estimated sugarcane yield, indicating that higher irrigation levels are linked to better yield. Conversely, minimum relative humidity shows a negative correlation (-0.31), suggesting that lower minimum humidity is associated with reduced yields. Among climatic covariates, maximum evapotranspiration correlates positively (0.59) with seasonal irrigation. In contrast, maximum daily temperature and maximum relative humidity exhibit a strong negative correlation (-0.75), indicating that higher minimum temperatures correspond to significantly lower relative humidity levels.

Modeling results

Table 2 presents a quantitative comparison between traditional statistical baselines (linear regression and Elastic Net) and ML models for sugarcane yield prediction. The results show that tree-based ensemble methods consistently outperform linear and penalized regression models across all evaluation metrics. In particular, CatBoost achieves the best predictive performance, with the highest coefficient of determination ($R^2 = 0.68$) and the lowest prediction error (RMSE = 10.8 t/ha, NRMSE = 0.72), indicating a superior ability to capture nonlinear relationships between climatic, soil, and management variables.

In contrast, linear regression and Elastic Net exhibit substantially lower explanatory power and higher prediction

errors, reflecting their limited capacity to model complex interactions and spatial variability in the data. Random forest and XGBoost yield intermediate performance, achieving comparable accuracy but remaining slightly inferior to CatBoost.

For CatBoost, a thorough hyperparameter tuning was conducted using grid search (Adnan et al. 2022) and fivefold cross-validation to optimize its performance. Key parameters included 500 iterations to strike a balance between complexity and training time, RMSE as the evaluation metric, and early stopping rounds set to 50, which halted training when no significant improvements were observed. The learning rate was set to 0.1 for controlled weight updates, and leaf regularization was configured at 3.0 to reduce overfitting and enhance generalization.

We provide a detailed view of the performance metrics obtained by CatBoost in Fig. 4. As expected for a flexible ensemble model, the model achieves a very high R^2 on the training set, indicating a strong fit. On the test set, the RMSE increases relative to the training set, reflecting a reduction in predictive accuracy on unseen data but suggesting reasonable generalization. Similarly, the NRMSE indicates lower relative accuracy on the test set compared to the training set. The coefficient of determination decreases to $R^2 = 0.68$ on the test set, indicating moderate to strong generalization performance and remaining higher than that of the baseline models. Overall, CatBoost achieves the best test performance among the evaluated methods, with an RMSE of 10.8 T/ha and $R^2 = 0.68$, indicating its robustness and superior accuracy in this study.

Covariates explainability

The covariates' importance in Fig. 5 highlights seasonal irrigation as the most critical factor influencing sugarcane yield, with significantly higher importance than other covariates. Minimum relative humidity and solar radiation follow with moderate influence, while precipitation and minimum temperature are less impactful. The latter underscores water management, particularly irrigation, and humidity, as key

Table 2 Quantitative comparison of baseline statistical models and ML models for sugarcane yield prediction. Performance is reported on the test set

Model	R^2	RMSE (t/ha)	NRMSE
Linear Regression	0.40	14.0	0.89
Elastic Net	0.32	15.0	0.95
Random Forest	0.65	11.5	0.83
XGBoost	0.65	11.2	0.80
CatBoost	0.68	10.8	0.72

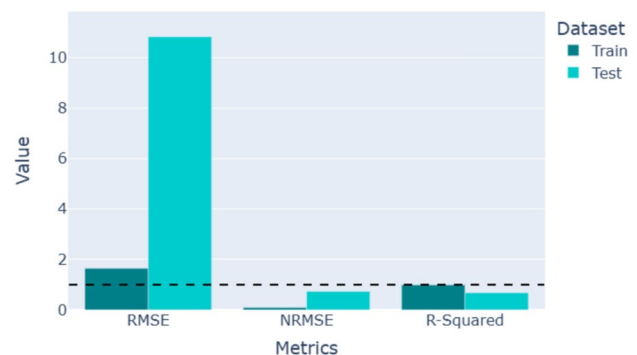
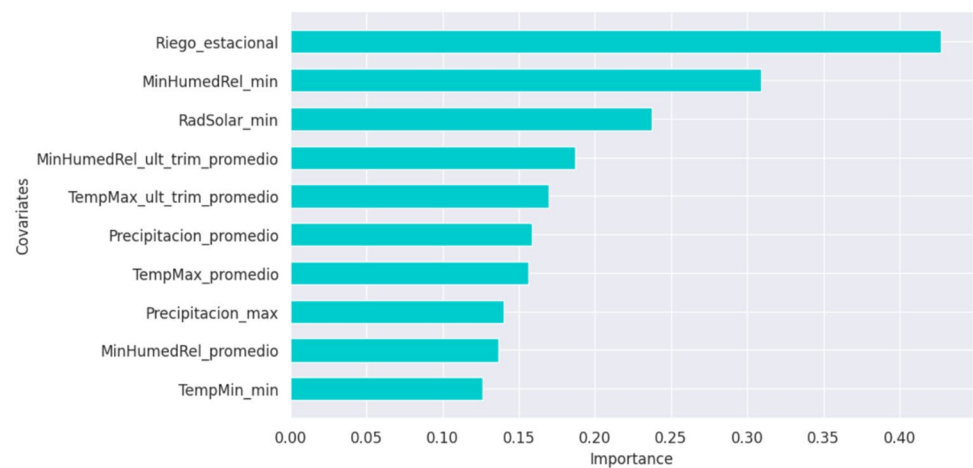


Fig. 4 Goodness-of-fit metrics for train and test datasets

Fig. 5 Covariates importance for CatBoost



determinants in yield prediction, with precipitation playing a secondary role.

Spatial prediction using ML models

Figure 6 shows the predictions using the CatBoost algorithm, identified as the most accurate in this study, which predicts sugarcane yields ranging from 91 to 110 T/ha, with higher-yield areas (100–110 T/ha) shown in purple and blue

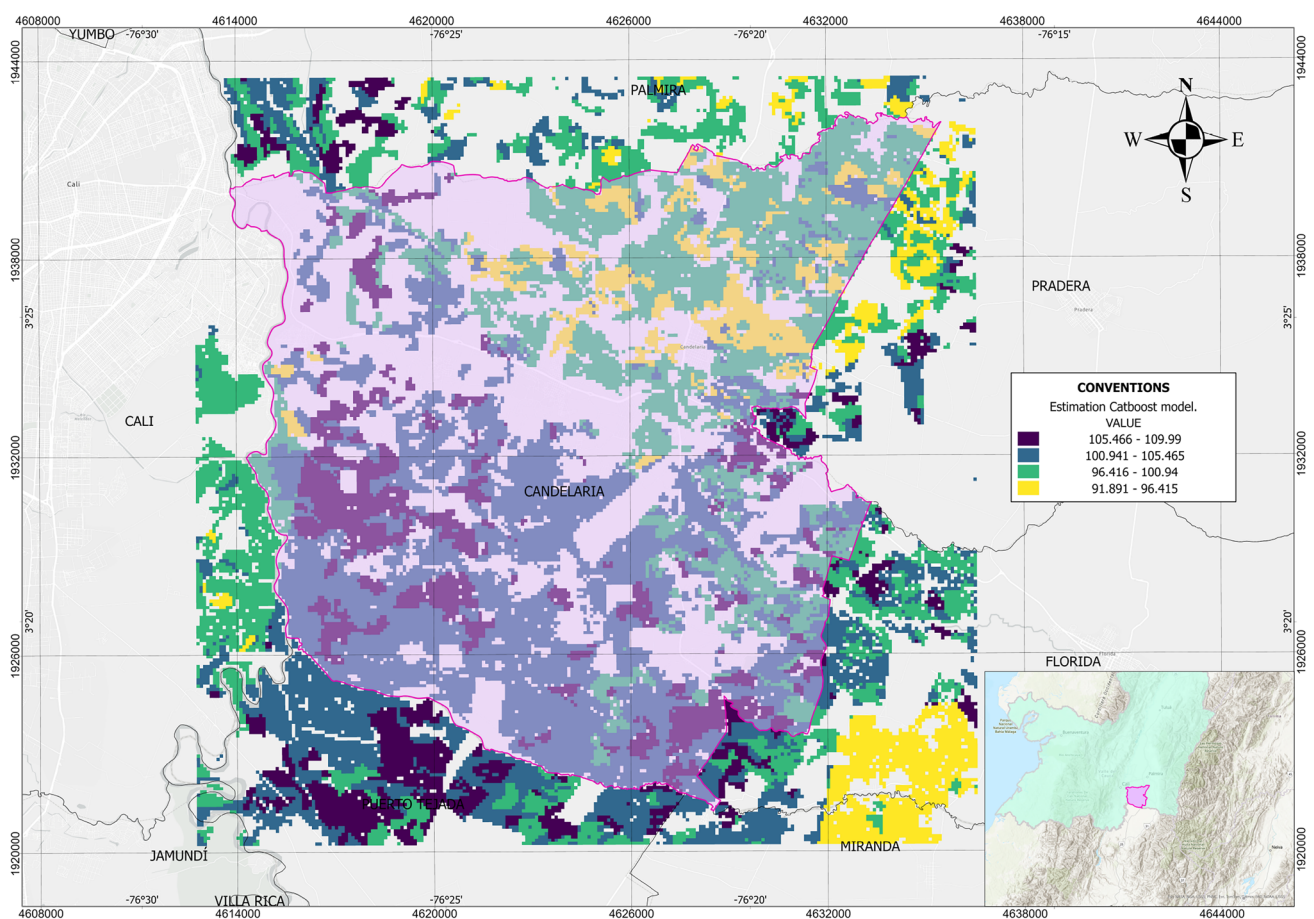


Fig. 6 Sugarcane estimation using CatBoost model

and the lower-yield regions (91–100 T/ha) in green and yellow. While CatBoost captures complex patterns and provides better generalization on new data compared to traditional methods, its predictions lack the intensity of AquaCrop's simulation. Despite this slight precision error, the model offers more detailed spatial variability, highlighting subtle differences in productivity across regions.

Figure 8 in Supplementary Material highlights areas where the CatBoost model underestimates sugarcane yield (purple, 2–6 T/ha) and overestimates it (yellow, 2–8 T/ha) compared to AquaCrop, while green areas show close alignment with AquaCrop predictions (–2 to 2 T/ha). Although CatBoost demonstrates superior accuracy overall, some discrepancies remain in specific regions. These differences are relatively small and localized, indicating that the model remains a robust tool for large-scale agricultural planning.

It is important to note that the comparison between CatBoost- and AquaCrop-generated yield maps does not constitute an independent validation, as both approaches rely on the same underlying input data and modeling assumptions. Instead, this comparison serves as an internal consistency check, illustrating how closely the ML model reproduces the spatial patterns encoded by the process-based simulator under identical conditions.

Discussion

While traditional models such as linear regression are often effective when considering specific factors like water availability and relative humidity, the results obtained with tree-based ML algorithms (CatBoost, XGBoost, and random forest) demonstrate a greater capacity to capture complex interactions among multiple climatic and hydric covariates. These findings are consistent with recent studies by dos Santos Luciano et al. (2021), who argue that ML models outperform traditional approaches in scenarios characterized by high spatial and temporal variability.

The results obtained in this research suggest that ML, combined with spatial data, can significantly improve sugarcane yield prediction in the municipality of Candelaria. This finding is consistent with previous studies that demonstrate the effectiveness of ML approaches in agricultural contexts, particularly for crop yield prediction. For example, Akbarian et al. (2024) and Rodríguez et al. (2019) report improved predictive accuracy when environmental and spatial variables are jointly considered.

It is essential to clearly define the nature of the learning task addressed in this study. The ML models developed here do not learn field-level yield variability directly from empirical harvest observations; instead, they learn the functional behavior of the AquaCrop crop simulation model under varying climatic, soil, and management conditions.

Accordingly, the proposed ML framework can be interpreted as a data-driven surrogate (or emulator) of AquaCrop, capturing its input–output relationships while enabling faster computation and higher spatial resolution.

This modeling choice has direct implications for real-world deployment. Currently, the trained ML models should be viewed as tools for scenario analysis, spatial exploration, and decision support under the assumptions encoded in AquaCrop, rather than as direct replacements for models trained on observed yields. Their predictions reflect physiologically consistent responses to irrigation, climate, and soil variability, which is particularly advantageous in data-scarce environments.

A limitation of this study is that the sugarcane yield used for model training was generated using the AquaCrop simulation model rather than measured at the field level, due to restricted access to empirical yield data from sugar mills in Valle del Cauca. Consequently, the proposed framework learns physiologically consistent yield responses encoded in the crop model but does not capture unmodeled sources of real field variability, such as pest pressure, management inefficiencies, or measurement noise, which limits its direct operational deployment. Nevertheless, AquaCrop has been extensively validated across diverse agroecological contexts (Coudron et al. 2023; Garcia-Vila et al. 2019; Sajid et al. 2022) and has been specifically calibrated for sugarcane in Valle del Cauca following FAO/DNP guidelines (Food 2013), supporting its use as a reliable proxy in data-scarce environments. Future work should incorporate empirical field-level yield data to recalibrate and externally validate the machine learning models, enabling a transition from surrogate modeling toward fully data-driven predictive systems.

Another limitation concerns the lack of explicit assessment of spatial autocorrelation in the data and model residuals. Because observations are spatially indexed on a regular grid, random train–test splits and cross-validation may lead to optimistic performance estimates due to spatial dependence between neighboring samples. Although tree-based ensemble models can implicitly capture some spatial structure through covariates and nonlinear interactions, they do not explicitly model residual spatial dependence. Future research should therefore adopt spatially blocked cross-validation, residual spatial diagnostics, or hybrid approaches combining machine learning with spatial statistical methods to more rigorously assess model generalization. Finally, the framework was applied to a single municipality and a single climatic year, limiting direct spatial and temporal extrapolation. Extending the analysis to multiple years and additional regions would allow the model to capture interannual climate variability and broader agroecological gradients, improving robustness and transferability.

Conclusions and future work

The spatial ML model developed in this work achieved good accuracy in predicting sugarcane yield in the Valle del Cauca by integrating geospatial and climatic data with algorithms like random forest, CatBoost, and XGBoost. The model emphasized irrigation as a key factor in optimizing water use and enhancing agricultural efficiency while addressing sustainability challenges. With an R-squared of 68% and RMSE of 10.8 T/ha, CatBoost outperformed traditional methods like linear regression and Elastic Net in capturing spatial variability in production factors. The approach successfully identified high- and low-yield areas, demonstrating the potential for adaptation to other regions and crops to support agricultural decision-making.

Future work should focus on extending the proposed framework along three complementary directions. First, incorporating empirical field-level yield data from sugarcane producers would enable the recalibration and external validation of the ML models, thereby reducing uncertainty originating from crop simulation and improving operational applicability. Second, expanding the analysis to multiple municipalities and multiple climatic years would allow the framework to capture broader agroecological gradients and interannual climate variability, thereby enabling a more rigorous assessment of spatial and temporal transferability. Third, future studies should adopt spatially explicit validation strategies, such as spatially blocked cross-validation and residual spatial autocorrelation diagnostics, to better quantify generalization performance and mitigate spatial leakage. Together, these extensions would strengthen the robustness, interpretability, and practical relevance of the proposed methodology for real-world sugarcane production systems.

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